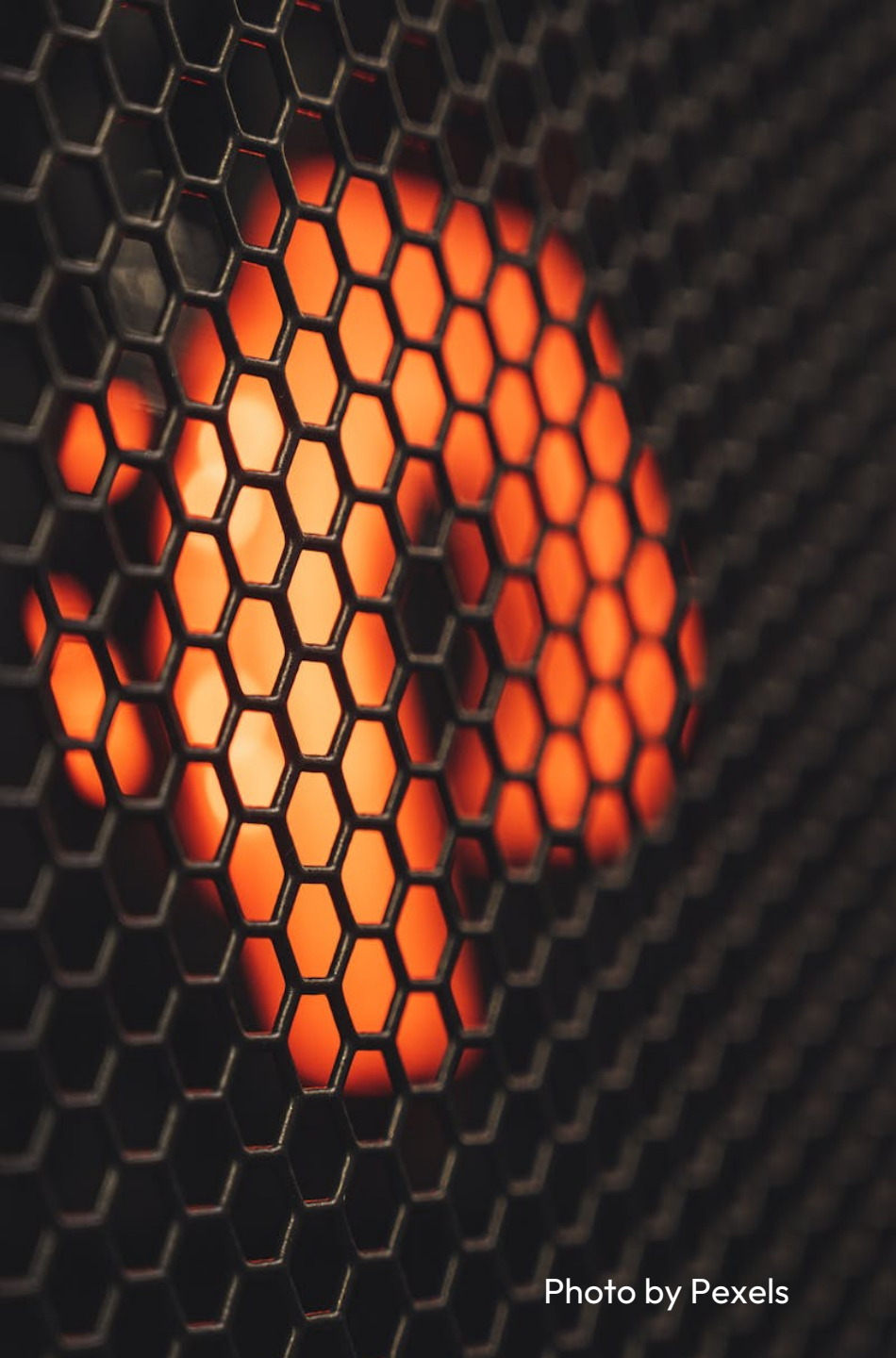




AUTOMATIC ATTENDANCE SYSTEM

Project Guide - Dr. I. Govardhana Rao Team- D.Keshav Sai Rao
Kamlekar Sai Siddharth Swargam Ramchandar Rao

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Introduction to the Project

Project Overview

- This project aims to develop an automatic attendance system using facial recognition technology.
- It addresses limitations of traditional manual sign-in sheets or instructor verification.
- The system is designed for educational institutions or workplaces to streamline attendance recording.
- Leveraging technology can make attendance taking more efficient and less error-prone.

Key Technologies

Main Components

- The project uses `dlib_face_recognition_resnet_model` for real-time face detection and recognition.
- Firebase serves as the backend for secure data storage and management in the system.
- Firebase Realtime Database is used to store student information and attendance records.
- Firebase Storage can optionally store student images for reference during recognition.



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Face Recognition Model

Model Details

- The `dlib_face_recognition_resnet_model` offers efficiency and scalability for large datasets.
- It is a robust model for real-time face detection and recognition in various conditions.
- This model supports the system's ability to accurately identify student faces in video frames.
- Its application ensures quick and reliable face recognition, improving overall system performance.

Firestore Integration

Backend Setup

- Firestore integration helps in secure data storage and management for the system.
- The Firestore Realtime Database stores essential student information and attendance records.
- Firestore Storage can be used to store student images for enhanced recognition accuracy.
- This integration ensures a seamless backend operation, maintaining data integrity and accessibility.

System Functionality

How it Works

- Live video is captured from a webcam for real-time processing in the system.
- The face recognition model identifies student faces within each video frame accurately.
- Upon recognition, student information is retrieved from the Firebase Realtime Database.
- The system updates the student's attendance record if a time threshold has passed since last attendance.

Visual Interface

User Interaction

- The system provides visual feedback on recognition success, attendance status, and potential errors.
- It displays student information like name, total attendance, major, year, and standing.
- Different modes (loading, success, attendance wait) are integrated for user interaction.
- The interface ensures users can easily understand and interact with the system effectively.

EncodeGenerator.py



Encoding Faces

- This script generates face encodings for student images and stores them with IDs in a pickle file.
- It helps create a reference for accurate face recognition in the system.
- The script also uploads student images to Firebase Storage if needed for future reference.
- Ensuring accurate encoding is crucial for the system's recognition efficiency.

AddDatatoDatabase.py



Database Integration

- This script populates the Firebase Realtime Database with pre-defined student information.
- Information includes name, major, starting year, total attendance, standing, and last attendance time.
- The script ensures all necessary data is available for the system's operation.
- Accurate database integration is key for the system to retrieve information efficiently.

Main Script



Core Functions

- The main script likely named `Face_Attendance.py` implements the system's core functionalities.
- It handles continuous video capture, face recognition, and attendance record updates in real-time.
- The script integrates all components to provide a seamless user experience.
- It ensures the system operates smoothly and efficiently during attendance taking.

Conclusion

Project Impact

- The Automatic Attendance System offers a modern solution to traditional attendance recording methods.
- It leverages advanced technologies to provide an efficient, error-free system.
- The project demonstrates the potential of integrating facial recognition and database management.
- It aims to significantly improve attendance management in educational institutions and workplaces.